

Practice Long Quiz 1

Logic, Proofs, and Basic Set Theory

Fall 2026

Name: _____ Student ID: _____

50 questions $\times$ 2 points = 100 points. Answer every question in its solution box. Show reasoning, intermediate work, or a counterexample as requested. Each entire numbered question is worth 2 points, including any subparts. For proofs, state assumptions and justify the conclusion; checking examples alone is insufficient.

Notation. $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ denote the integers, rationals, and reals. $\mathcal{P}(A)$ is the power set of A ; $A^c = U \setminus A$ is the complement relative to the stated universe U . $A \subseteq B$ allows equality; $A \subsetneq B$ means a proper subset. The symbol $\vee$ is inclusive OR.

A. Logic – Questions 1–20

40 points

Question 1

[2 points]

Which of the following are propositions? Briefly justify each answer: (a) “7 is a prime number”; (b) “ $x + 2 = 5$,” where x is an unspecified integer.

Solution

Question 2

[2 points]

Let p mean “I submit the assignment” and q mean “The assignment is accepted.” Translate into propositional logic: (a) “The assignment is accepted only if I submit it”; (b) “I submit the assignment, but it is not accepted.”

Solution

Question 3**[2 points]**

Suppose p is true, q is false, and r is true. Find the truth value of $(p \wedge \neg q) \rightarrow (q \vee \neg r)$. Show the truth values of the antecedent and the consequent.

*Solution***Question 4****[2 points]**

Write the converse, inverse, and contrapositive of the statement: “If an integer is divisible by 6, then it is even.”

*Solution***Question 5****[2 points]**

Construct a complete truth table for $p \leftrightarrow \neg q$. Include columns for p , q , $\neg q$, and the final expression.

Solution

Question 6**[2 points]**

Use a truth table to determine whether $(p \wedge q) \rightarrow p$ is a tautology, a contradiction, or a contingency.

*Solution***Question 7****[2 points]**

Simplify $\neg[p \vee (\neg p \wedge q)]$ using laws of logic. Name a law used at each step. Do not use a truth table.

*Solution***Question 8****[2 points]**

Use logical equivalence laws to show that $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$. Name the laws you use.

Solution

Question 9**[2 points]**

Show that $p \rightarrow q$ and $q \rightarrow p$ are not logically equivalent. Give one truth assignment and evaluate both expressions under it.

*Solution***Question 10****[2 points]**

Rewrite $\neg(p \leftrightarrow q)$ in disjunctive normal form (an OR of ANDs of variables or their negations). Show your equivalence steps.

*Solution***Question 11****[2 points]**

Let S be a set of students, B a set of books, and $R(s, b)$ mean “student s has read book b .” Express “Every student has read at least one book” using quantifiers. Specify the domain of each variable.

Solution

Question 12**[2 points]**

Let D be a set of students, and let $A(x)$ mean “ x is absent.” Express “Exactly one student is absent” using quantifiers, equality, and logical connectives. Do not use the symbol $\exists!$.

*Solution***Question 13****[2 points]**

Negate $\forall x \in \mathbb{R} \exists y \in \mathbb{R} (x + y = 0)$. Push the negation through every quantifier and express the result using $\neq$, with no negation symbol in front of a quantifier.

*Solution***Question 14****[2 points]**

Determine the truth value of each statement and justify your answers: (a) $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (y > x)$; (b) $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (y > x)$.

Solution

Question 15**[2 points]**

Determine whether $\exists x (x^2 = 2)$ is true when the domain is (a) $\mathbb{Z}$ and (b) $\mathbb{R}$. Justify each answer.

*Solution***Question 16****[2 points]**

From the premises $p \rightarrow q$, $q \rightarrow r$, and p , derive r . Give a short sequence of statements and name the inference rule used at each step.

*Solution***Question 17****[2 points]**

Is the argument with premises $p \rightarrow q$ and q , and conclusion p , valid? If valid, justify it; if invalid, give a truth assignment making both premises true and the conclusion false.

Solution

Question 18**[2 points]**

All variables range over students. From $\exists x T(x)$ and $\forall x (T(x) \rightarrow P(x))$, derive $\exists x P(x)$. Explain how you use a witness for the existential premise.

*Solution***Question 19****[2 points]**

Can $p \vee q$, $\neg p \vee r$, $\neg q$, and $\neg r$ all be true at once? Give a satisfying truth assignment or explain why none exists.

*Solution***Question 20****[2 points]**

Negate $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} \forall z \in \mathbb{Z} (x + y > z)$. Push the negation through all three quantifiers and express the result using $\leq$.

Solution

B. Proofs – Questions 21–40**40 points****Question 21****[2 points]**

Give a direct proof that the sum of two odd integers is even. Start from the definition of an odd integer.

*Solution***Question 22****[2 points]**

Give a direct proof that the product of two rational numbers is rational. State why the denominator in your resulting fraction is nonzero.

*Solution***Question 23****[2 points]**

Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. Prove directly that if $a \mid b$ and $a \mid c$, then $a \mid (3b - 2c)$. Use the definition of divisibility.

Solution

Question 24**[2 points]**

Prove by contraposition that, for every $n \in \mathbb{Z}$, if n^2 is even, then n is even. State the contrapositive explicitly.

*Solution***Question 25****[2 points]**

Let $m, n \in \mathbb{Z}$. Prove by contraposition that if $m + n$ is odd, then at least one of m and n is odd.

*Solution***Question 26****[2 points]**

Prove by cases that $3 \mid n(n+1)(n+2)$ for every $n \in \mathbb{Z}$. Use the three cases $n = 3k$, $n = 3k + 1$, and $n = 3k + 2$.

Solution

Question 27**[2 points]**

Prove by contradiction that there is no greatest rational number.

*Solution***Question 28****[2 points]**

Let r be rational and x be irrational. Prove by contradiction that $r + x$ is irrational. You may use the fact that the difference of two rational numbers is rational.

*Solution***Question 29****[2 points]**

Disprove the claim: “For every integer $n \geq 0$, the number $n^2 + n + 41$ is prime.” Give a counterexample and a factorization showing that its value is composite.

Solution

Question 30**[2 points]**

Prove that, for every $n \in \mathbb{Z}$, n is even if and only if $3n + 2$ is even. Clearly establish both directions.

*Solution***Question 31****[2 points]**

A student writes: “Let m, n be odd integers. Then $m = 2k + 1$ and $n = 2k + 1$ for some integer k , so $m + n = 4k + 2$ is even.” Explain the unjustified assumption and repair the proof.

*Solution***Question 32****[2 points]**

Prove by mathematical induction that $\sum_{i=1}^n (2i - 1) = n^2$ for every integer $n \geq 1$. Include the base case, induction hypothesis, and inductive step.

Solution

Question 33**[2 points]**

Prove by mathematical induction that $3 \mid (4^n - 1)$ for every integer $n \geq 0$. Include the base case and the inductive step.

*Solution***Question 34****[2 points]**

Complete a strong-induction proof that every integer $n \geq 2$ is a product of primes (a single prime counts). The base case $n = 2$ holds. State the strong induction hypothesis and prove the step for $n = k + 1$ by treating the prime and composite cases.

*Solution***Question 35****[2 points]**

A student claims $2^n \geq n + 2$ for all integers $n \geq 0$, using only: “If $2^k \geq k + 2$, then $2^{k+1} \geq 2k + 4 \geq k + 3$.” Explain why this is not a proof of the stated claim. Find the smallest starting value of n for which this step and a valid base case prove it.

Solution

Question 36**[2 points]**

Prove by mathematical induction that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for every integer $n \geq 1$. Include the base case, induction hypothesis, and inductive step.

*Solution***Question 37****[2 points]**

Prove by mathematical induction that $2^n > n$ for every integer $n \geq 1$.

*Solution***Question 38****[2 points]**

Prove by mathematical induction that $6 \mid (n^3 - n)$ for every integer $n \geq 0$. Include the base case and the inductive step.

Solution

Question 39**[2 points]**

Prove by mathematical induction that a set with n elements has exactly 2^n subsets, for every integer $n \geq 0$. In the inductive step, split the subsets of a set with $k + 1$ elements into those that contain the new element and those that do not.

*Solution***Question 40****[2 points]**

Use strong induction to prove that every integer $n \geq 2$ can be written as a sum of 2s and 3s (one or more terms, each equal to 2 or 3). State the base cases needed and complete the inductive step for $k + 1 \geq 5$.

Solution

C. Basic Set Theory – Questions 41–50**20 points****Question 41****[2 points]**

Write the set $\{x \in \mathbb{Z} \mid -3 \leq x < 4 \text{ and } x^2 \leq 4\}$ in roster form, and give its cardinality.

*Solution***Question 42****[2 points]**

Let $A = \{\emptyset, \{1\}, 1\}$. Determine whether each statement is true or false, with a brief reason: (a) $\emptyset \in A$; (b) $\{1\} \subseteq A$; (c) $\{\emptyset\} \in A$; (d) $\{\{1\}\} \subseteq A$.

*Solution***Question 43****[2 points]**

Are $\emptyset$ and $\{\emptyset\}$ equal? Give the cardinality of each set and explain your answer.

Solution

Question 44**[2 points]**

Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$, and $C = \{3, 2, 1\}$. Determine whether $A \subseteq B$, $A \subsetneq B$, and $A = C$ are true or false. Briefly justify each answer.

*Solution***Question 45****[2 points]**

List every element of $\mathcal{P}(\{a, b\})$, where $a \neq b$. Use braces carefully to distinguish elements from subsets.

*Solution***Question 46****[2 points]**

A finite set A has exactly 32 subsets. Find $|A|$ and the number of proper subsets of A . Show the counting rule you use.

Solution

Question 47**[2 points]**

Let $A = \{1, 2, 4, 6\}$ and $B = \{2, 3, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

*Solution***Question 48****[2 points]**

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 4, 6\}$, and $B = \{2, 3, 6, 7\}$. Find $A \setminus B$ and A^c .

*Solution***Question 49****[2 points]**

Let $A = \{1, 2, 4\}$ and $B = \{2, 3, 4, 5\}$. Compute the symmetric difference $A \triangle B = (A \setminus B) \cup (B \setminus A)$. Show the two differences before taking their union.

Solution

Question 50**[2 points]**

In a group of 40 students, 24 study computer science, 21 study mathematics, and 10 study both. How many study at least one of the two subjects? How many study neither? Show your calculation.

Solution